



Application Design Specifications

Mexicana - Appointment Scheduling Application.doc

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Table of Contents

Table of Contents.....	1
1 Introduction.....	1
1.1 Purpose of Document.....	1
1.2 Intended Audience.....	2
1.3 Scope.....	2
1.4 Terminology.....	2
1.5 Document Structure.....	2
1.6 Related Documents.....	2
2 Overview.....	3
2.1 General.....	3
2.1.1 Inbound - Appointment Scheduling Agent Module:.....	3
2.1.2 Inbound - Appointment Re-Scheduling Executive Agent Module:.....	3
2.1.3 Outbound - Client Solicitation Agent Modules:.....	3
2.1.4 Outbound - Appointment Confirmation Agent Module:.....	4
2.1.5 Outbound - Visit Confirmation Agent Module:.....	4
2.1.6 Web Update - Agent Assignment Executive Agent Module:.....	4
2.1.7 Web Update - Visit Outcome Agent Module:.....	4
2.2 Application Context.....	4
2.3 User Interface.....	4
2.4 Technologies Used.....	7
3 Process Breakdown.....	7
3.1 Logical Data Flow.....	7
3.2 Data Definitions.....	8
3.3 Components.....	8
3.4 Dependencies.....	8
4 Routine Breakdown.....	8
4.1 Introduction.....	8
4.2 Module Hierarchy.....	8
4.3 Module Descriptions.....	8

1 Introduction

1.1 Purpose of Document

This document is used to prototype an application. It will list modules, database structure, functions, processes, application features, GUI screens, technologies involved, work and data flow diagrams, user interactions, automatic processes, communications,



Application Design Specifications

Mexicana - Appointment Scheduling Application.doc

logging and notification capabilities, application limitations, time estimates and any foreseeable problems or recommendations.

It will also list configuration steps, application setup and routine maintenance if applicable.

1.2 Intended Audience

- ✓ Project Managers
- ✓ Programmers
- ✓ System Testers
- ✓ Database Administrators
- ✓ Telvista Management and/or Administration Personnel
- ✓ Partner Company Personnel involved with this application

1.3 Scope

This document provides detailed information on the application, modules, interfaces and database structure. However, because there may be other people involved in the design process anything may be subject to change. Changes should be coordinated and reflected in this document.

We should make any attempt to list any information that is relevant. Nevertheless there may be some omissions. We should make every attempt to minimize omissions but if something has to be omitted we should note it. This is so that we prevent time costly assumptions.

It is assumed that the intended audience is able to understand most of the technical terminology. Therefore, some details are omitted. While some terms are typically only understood by programmers or database administrators the definitions of those terms or explanations of those terms may be omitted so that the size of this document can remain manageable.

Only in those cases dimmed necessary will those terms or procedures be explained in more detail, non technical terms or at all. However, any members of the audience's contributions are necessary and welcome to the success of the application. Therefore, in those cases, they should ask the questions and we should try to clarify them.

1.4 Terminology

The word module is used loosely to mean either an application module which can correspond directly with a GUI screen or operation mode. Or, it could mean a function module which could also be a sub-module. This is to be determined based on the context.

1.5 Document Structure

Standard.

1.6 Related Documents

- [PowerPoint Presentation](#) with initial work flow diagram
- [Access Database](#) with database structure prototype

2 Overview

2.1 General

This call center application will be used to setup, track, and follow-up appointments. The appointments can be re-scheduled. The appointments are to be confirmed two days prior to the date. The record is updated once the appointment is met. In order to do this, the system calls the client and the agent will request information from the client.

The agents through a predictive dialing system receive inbound and outbound calls. *There are 2 inbound modules:*

2.1.1 Inbound - Appointment Scheduling Agent Module:

1. Adding a client or editing client information
2. Client screening and service approval
3. Appointment scheduling or re-scheduling
4. Prospect referral

2.1.2 Inbound - Appointment Re-Scheduling Executive Agent Module:

1. Appointment re-scheduling

There are 3 outbound modules:

2.1.3 Outbound - Client Solicitation Agent Modules:

This module has 3 modes of operation which determine in which database the data will be:

2.1.3.1 Frequent Flyer

2.1.3.2 Client Reference

2.1.3.3 Scheduled Calls

The modes of operation also affect how the agent will relate to the client as far as what will be said and how it will be said, but not how the work flow is handled. The work flow is the same, only the underlying database will be different. Actually, the database will be the same, but will contain a field to differentiate the mode of operation. It will only appear that way logically, structurally it will reside in the same table. This module will be used for:

1. Editing client (prospect) information
2. Client screening and service approval
3. Appointment scheduling
4. Prospect referral

2.1.4 Outbound - Appointment Confirmation Agent Module:

1. Verifying appointment
2. Re-scheduling appointments
3. Screening out clients who re-schedule more than x times. Canceling

2.1.5 Outbound - Visit Confirmation Agent Module:

1. Verifying visit
2. Polling client satisfaction and other information

There are 2 web update modules:

2.1.6 Web Update - Agent Assignment Executive Agent Module:

1. Searching database based on selected criteria
2. Assigning agent to visit client

2.1.7 Web Update - Visit Outcome Agent Module:

1. Searching database based on selected criteria
2. Update database: Appointment Status, Visit Status, Contract Information, Client Disqualification Information if applicable

There may also be a Reporting Module so that supervisors or administrative personnel can monitor activity, progress, numbers, etc...

2.2 Application Context

There are 3 possible contexts:

- C1 - Agent. Carries out various operations as mentioned above
- C2 - Executive Agent. Can assign agents. May view certain reports
- C3 - Administrative. Can view reports and change application parameters and data catalogs. To minimize development time the parameters and data catalogs can be changed from MS Access, by linking the actual tables wherever they may reside. Since they aren't likely to change that often. Or, we could design a web module for this purpose.

2.3 User Interface

The GUI will be a series of screens that correspond to the main modules. In other words, each application module will have its own screen. However, because the modules share similar tasks, some of these screens will be shared by the modules. In fact, some of the tasks are identical across the modules.

The application and the modules should attempt to make the work as easy, as fast and as pleasant as possible by automating and anticipating tasks.

User Modes:

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- M1- Inbound - Appointment Scheduling
- M2 - Inbound - Appointment Re-Scheduling
- M3 - Outbound - Client Solicitation - Frequent Flyer
- M4 - Outbound - Client Solicitation - Client Reference
- M5 - Outbound - Client Solicitation - Scheduled Calls
- M6 - Outbound - Appointment Confirmation
- M7 - Outbound - Visit Confirmation
- M8 - Web Update - Agent Assignment
- M9 - Web Update - Visit Outcome

GUI Screens:

-  S1 - Login Screen
 - a. Every employee must login
-  S2 - Module Selection Screen
 - a. Application presents a module selection screen based on employee's credentials
 - b. Every employee must select a screen
-  S3 - Call Disposition Update Screen
 - a. All calls must be logged and updated with a call disposition code
 - b. After update, this screen brings up another screen based on the user mode of operation
-  S4 - Client Search Screen - Used only by inbound module
 - a. In Appointment Scheduling Mode, this screen brings up Client Information Screen to edit or add client. Then, it can either schedule or re-schedule an appointment, so it brings up Appointment Setup Screen
 - b. In Appointment Re-Scheduling Mode, this screen brings up Appointment Setup Screen
-  S5 - Client Information Screen
 - a. Used to edit or add client information
 - b. Agent should ask client or lead for best times to call at the various phone numbers
 - c. Used to add client referral information. In this mode of operation some fields are hidden, such as credit card or earnings information
 - d. When finished updating this screen brings up another based on user mode
-  S6 -Appointment Setup Screen
 - a. Used to schedule or re-schedule appointments

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- b. It has date and time selection controls and it reflects if a day is unavailable for visits based on limitations setup by application administrator based on the city

S7 - Appointment Confirmation Screen

- a. Used to confirm, re-schedule, or cancel an appointment based on a limit on number of re-scheduling transactions

S8 - Agent Assignment Record Search Screen

- a. Search appointments based on diverse criteria so that supervisor or executive can assign an agent

S9 - Agent Assignment Screen

- a. View search results and assign an agent to visit a client

S10 - Visit Status Record Search Screen

- a. Search database for completed visits based on diverse criteria so that agent can update visit status and other information

S11 - Visit Status Update Screen

- a. View search results and update visit information and enter contract number if sale was realized

S12 - Visit Confirmation Screen

- a. Ask client questions about the visit

Processes:

- ☐ P1 – Responsible Party Determination
- ☐ P2 – Redial Scheduling ???
- ☐ P3 – Data Validation ???
- ☐ P4 – Disinterested Client Classification
- ☐ P5 – Disqualified Client Classification
- ☐ P6 – Failed Visit Classification
- ☐ P7 – Canceling Appointments

The following is a list of all operations and their relationships:

Name	Operation	Used In	GUI Type
S1	Agent login	All Modes	Text Fields
S2	Work module selection	All Modes	Graphics and Text
S3	Call classification	M1 – M7	List Box or Graphics
S4	Client search	M1, M2	Text Fields
S5	Client information	M1 – M7	List Boxes, Text Fields
S6	Appointment setup	M1 – M6	Text Fields, Date Controls
S7	Appointment confirmation	M6	Text Fields
S8	Agent assignment record search	M8	Text Fields

Application Design Specifications
Mexicana - Appointment Scheduling Application.doc

S9	Agent assignment	M8	List Boxes
S10	Visit status record search	M9	Text Fields
S11	Visit status update	M9	List Boxes
S12	Visit confirmation	M7	List Boxes, Text Fields
P1	Responsible party determination	S5, S6, S7, S12	List Box
P2	Redial scheduling	S5, S6, S7, S12	Date Control
P3	Data validation	S5	(Existing Fields)
P4	Uninterested client classification	S6	List Box
P5	Disqualified client classification	S6	List Box
P6	Failed visit classification	S12	List Box
P7	Canceling appointments	S7	Function - Automatic

2.4 Technologies Used

- Microsoft ASP.NET
- MS SQL Server
- Predictive Dialing System
- Multi Tier Architecture
- CSS – Cascading Style Sheets
- User Friendly Graphical User Interface

3 Process Breakdown

3.1 Logical Data Flow

See [PowerPoint Presentation](#) related document above for preliminary information.

For all Modes: S1 - Login □ S2 - Mode Selection

- M1 – Inbound - Appointment Scheduling
 - Call begins
 - S3 – Call classification: If invalid, update record, end
 - S4 – Client search: Obtain telephone number from caller id string, or ask client for acct. no. and search database. If not found add a new client
 - If client has a pending appointment, and if he wants to re-schedule go to S6 – Appointment Setup. If not re-scheduling add notes on call
 - If no pending appointment go to S5 – Client Information
 - If client disqualified, P5, end call



Application Design Specifications

Mexicana - Appointment Scheduling Application.doc

- o If client uninterested, P4, end call

□ M2 – Inbound - Appointment Re-Scheduling

□ M3, M4, M5 – Outbound - Client Solicitation

□ M6 – Outbound - Appointment Confirmation

□ M7 – Outbound - Visit Confirmation

□ M8 – Web Update - Agent Assignment

□ M9 – Web Update - Visit Outcome

3.2 Data Definitions

See [Access Database](#) related document above for complete details.

3.3 Components

3.4 Dependencies

4 Routine Breakdown

4.1 Introduction

4.2 Module Hierarchy

4.3 Module Descriptions